STATISTICS

Introduction to Statistics

Statistics is the discipline concerned with collection, organization, analysis, interpretation, and presentation of data.

Types of Data

Type	Meaning	Example
Categorical (Qualitative)	Names or Labels	Gender, City, Rank
Numerical (Quantitative)	Numbers	Age, Income, Marks
Discrete	Countable numbers	Number of students
Continuous	Measured values	Height, Weight

1. Levels of Measurement

- Nominal – No order (e.g., Names)

- Ordinal – Ordered categories (e.g., Rank)
- Interval – No true zero (e.g., Temperature in Celsius)

Ratio – True zero exists (e.g., Weight, Height)

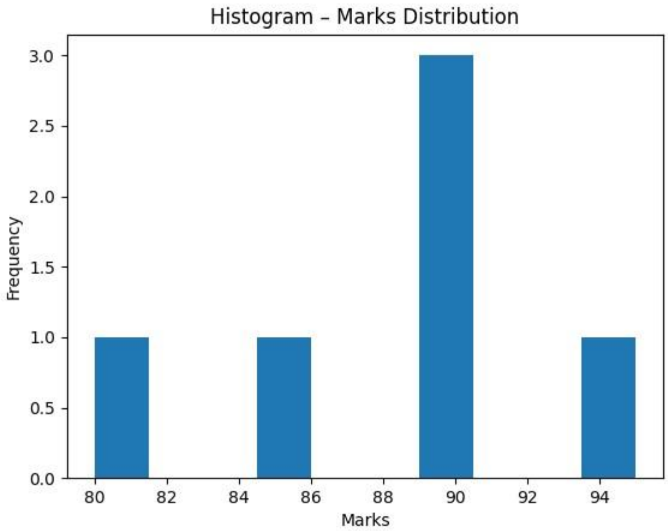
- Measures of Central Tendency

Data: [90, 85, 80, 90, 95, 90]

Mean = 88.33

Median = 90.0

Mode = 90 (Most frequent value)

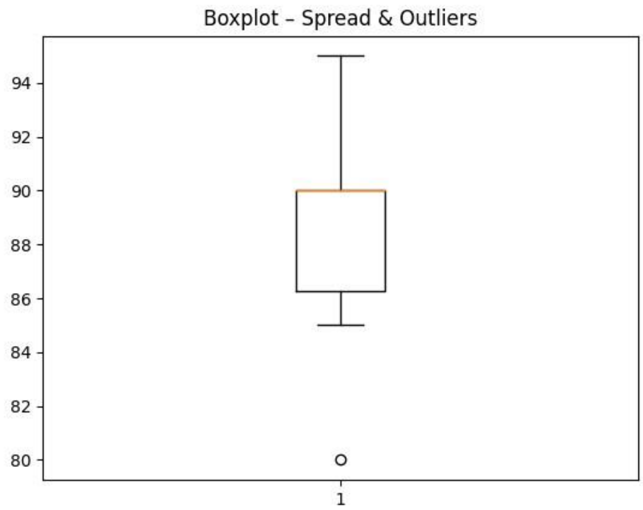


2. Measures of Dispersion

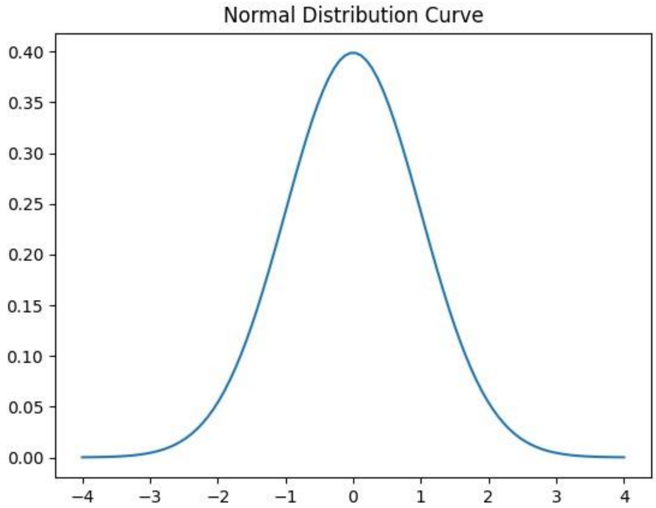
Variance = 22.22

Standard Deviation = 4.71

Standard deviation tells how much data deviates from mean.



3. Normal Distribution & Empirical Rule



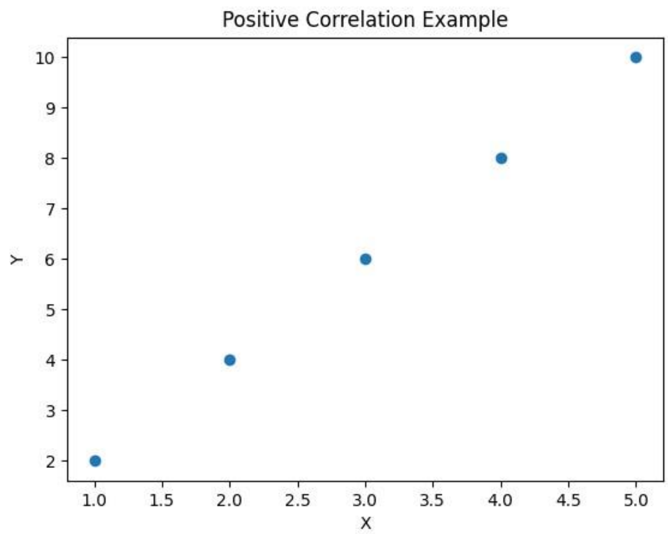
Empirical Rule: 68% within 1 SD, 95% within 2 SD, 99.7% within 3 SD.

Chebyshev Theorem: At least $(1 - 1/k^2)$ data lies within k SD.

4. Covariance & Correlation

Correlation Coefficient (r) = 1.0

Positive correlation means both variables move in same direction.



5. Percentiles & Quartiles

Q1 (25th Percentile) = 1739.5

Median (50th Percentile) = 2038.0

Q3 (75th Percentile) = 2151.0

6. Standardization (Z-Score)

Mean Age = 30.0

Standard Deviation = 1.41

Z-Scores = [-1.41 -0.71 0. 0.71 1.41]

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13. Sampling & Central Limit Theorem

Population = Entire dataset

Sample = Subset of population

CLT: Sample means form normal distribution as sample size increases.